Practical exercise 6

1. Enter the data in Table 1 into `R` using one of the usual ways. y is the response variable and x_1 and x_2 the explanatory variables.

y	x_1	x_2
3.5	3.1	30
3.2	3.4	25
3.0	3.0	20
2.9	3.2	30
4.0	3.9	40
2.5	2.8	25
2.3	2.2	30

Table 1

   (a) Combine all three vectors in a data frame called `illustration` using

   ```
   > illustration = data.frame(y, x1, x2)
   ```

   (b) Produce a 'matrix plot' of scatterplots for pairs of variables using

   ```
   > pairs(illustration)
   ```

   (c) Try to find a linear relationship between the three variables using

   ```
   > model1 = lm(y~x1+x2)
   ```

   Write down the equation of the fitted model. Use the `summary` command to obtain the residual standard error and the standard errors of $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$.

   (d) Produce plots of the residuals against x_1, x_2 and the fitted values $\hat{y}$ in turn using

   ```
   > par(mfrow=c(2,2))
   > plot(residuals(model1)~x1)
   > plot(residuals(model1)~x2)
   > plot(residuals(model1)~fitted.values(model1))
   ```

   Return the plotting window to normal using

   ```
   > par(mfrow=c(1,1))
   ```

   (e) Examine the contribution made by both x_1 and x_2 to the total variation in the responses using the command

   ```
   > summary.aov(model1)
   ```

   (f) Use the `R` output to perform an F-test for the overall fit of the model and explain your conclusion.

2. The data frame `trees` is made available in `R` with

```
> data(trees)
```

The data set records the girth in inches, height in feet and volume of timber in cubic feet of each of a sample of 31 felled black cherry trees in Allegheny National Forest, Pennsylvania. Note that girth is the diameter of the tree (in inches) measured at 4 ft 6 in above the ground.

Treating volume as the (continuous) response variable y, try to find a reasonable model describing its distribution conditional on the explanatory variable height as follows.

Obtain all possible scatter diagrams of the data using the command

```
> pairs(trees)
```

First try to fit a linear model using

```
> model1=lm(Volume~Height)
```

Also make reference to influence by using Cook's Distance. You might want to try the commands

```
> plot(cooks.distance(model1)~Height)
> plot(model1, which=4)
```

and observe the difference in the output.

We might now ask if we can find a model with both explanatory variables height and girth. Physical considerations suggest that we should explore the very simple model

$$\text{Volume} \quad = \quad b_1 \times \text{height} \times (\text{girth})^2 + \epsilon$$

This is basically the formula for the volume of a cylinder.

You can fit the model by using the `R` command

```
> model2=lm(Volume~I(Height*Girth^2)-1)
```

Again comment on its appropriateness by making reference to standard errors and residuals.

Note: for a model without intercept, `R` computes the value of R^2 using a different formula than for models with intercept. Type `?summary.lm` and take a look at the formula given for `r.squared` – in particular, notice how it changes if there is no intercept. Try re-fitting the above model with an intercept present; notice that the model must be a better fit to the data (we've removed the constraint that intercept must be zero), but that the computed R^2 value decreases.

3. The data frame `whiteside` which deals with gas consumption is made available by:

```
> library(MASS)
> data(whiteside)
```

It records the weekly gas consumption and average external temperature at a house in south-east England during two heating seasons, one before and one after cavity-wall insulation was installed. The variables are:

Variable	**Description**
Gas	weekly gas consumption
Temp	average external temperature during week
Insul (binary factor)	Before (insulation) or After

We regard gas consumption as the response variable and seek to model its dependence on the explanatory variables temperature and insulation.

(a) Plot the data, using different symbols to distinguish gas consumption before and after insulation.

(b) Separately for each of the two values of insulation, investigate the dependence of gas consumption on temperature. Use graphical techniques, fit appropriate linear models, and investigate their adequacy. Note that `R` code to fit linear models to subsets of the data is of the form:

```
> attach(whiteside)
> gas.before = lm(Gas ~ Temp, subset = Insul=="Before")
> gas.after = lm(Gas ~ Temp, subset = Insul=="After")
```

Examine the summaries of the two models.

(c) We now want to include both covariates in the model – there are several options. The plot of the data suggests that the fitted lines for gas consumption before and after insulation could have separate intercepts. One way to fit this model in `R` is by using:

```
> gas.sep.intercept = lm(Gas ~ Insul + Temp )
> summary(gas.sep.intercept)
```

Examine the summary and write down the fitted model.

(d) We can also fit a model with different slopes for before and after insulation:

```
> gas.sep.slope = lm(Gas ~ Temp:Insul)
> summary(gas.sep.slope)
```

Again, look at the summary and write down the fitted model.

(e) Finally, we can try a model which gives different intercept and slope:

```
> gas.sep.interc.slope = lm(Gas ~ Insul * Temp)
> summary(gas.sep.interc.slope)
```

Note that you can add the fitted lines to your plot using:

```
>lines(fitted(gas.sep.interc.slope)[Insul=="Before"]~Temp[Insul=="Before"])
>lines(fitted(gas.sep.interc.slope)[Insul=="After"]~Temp[Insul=="After"],col="red")
```

(You need to have plotted the data first, as in (a) above.)
Compare the results of this with your earlier analyses.

(f) Use the results of your analyses to assess the importance of insulation and write a short report summarising clearly your findings.

4. The data `twins` consist of IQ scores for identical twins, one raised by foster parents, the other by the natural (biological) parents. We also know the social class of the natural parents (high, middle, or low).

   We are interested in predicting the IQ of the twin with foster parents from the IQ of the twin with the natural parents and the social class of natural parents.

   The `R` data frame `twins` is available on the course web page on VISION. Read in the data using `dget`. Use code similar to that in Q3 to do the following:

   (a) Plot the Foster against the Biological twin IQ with the natural parents' social class indicated on the graph. Based on the plot suggest an appropriate model (e.g. inclusion of separate intercept and/or slope, interaction term).

   (b) Fit the most general possible model, i.e. the model including separate intercepts and slopes (interaction term). Does the model seem appropriate?

   Based on the fitted model, write down an expression that will give you the predicted IQ of a fostered twin when the social class of the biological parents is 'low' and the IQ of the twin raised by biological parents is 102.

   Calculate this predicted value and verify your answer using the `predict()` command in `R`.

   Does the analysis suggest that there is interaction between the covariates (biological twin IQ and class)?

   (c) Now fit a model with different intercepts but the same slope for all 3 social classes (this is a 'parallel lines' model).

   Use the fitted model to predict again the IQ of a foster twin when the social class of the biological parents is 'low' and the IQ of the biological twin is 102.

   Which of the two models (in parts (b) and (c)) would you choose for this analysis?

   (d) Use the `anova()` command on your first model (in part (b)). The output compares the 3 possible sub-models:

   $$\text{Foster} \sim \text{Biological}$$
   $$\text{Foster} \sim \text{Biological} + \text{Social}$$
   $$\text{Foster} \sim \text{Biological} + \text{Social} + \text{Biological:Social}$$

   It compares them by performing appropriate F-tests for sequentially including the corresponding additional covariate, and giving relevant p-values.

   Based on the `anova()` output, comment on whether the model in part (c) can be further simplified (and how).